SOFTWARE ARCHITECTURE AND DESIGN SPECIFICATION

ONLINE QUIZ SYSTEM

PHASE 1 – VALIDATION SPECIFICATION | TEAM 13

Document Information

Item	Details
System	Online Quiz System
Department	CSE
Subject	Software Engineering
Semester / Section	5TH SEM, H
Team	13
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1. Introduction

The Online Quiz System is a web-based system used to conduct multiple-choice quizzes. A Quiz Participant can start a quiz, view questions, select answers, submit the quiz, and view the resulting score. An Administrator can log in and manage quiz questions by adding, editing, deleting, and viewing questions.

2. System Actors and Responsibilities

Actor	Description	Use Cases
Quiz Participant/User	The participant takes the online quiz, answers multiple-choice questions, navigates between questions, submits the quiz, and views the final score and result.	Start Quiz; View Question; Select Answer; Navigate Questions; Submit Quiz; View Score; View Result; Exit Quiz
Administrator	The administrator manages quiz questions and must log in using valid credentials before accessing administrative functions.	Login; View Questions; Add Question; Edit Question; Delete Question

3. Architecture Information Available in the Source

The source document describes the system's user-facing functions, administrator functions, validation behaviour and use-case relationships. It does not specify a software architecture pattern or provide a component diagram and component descriptions.

3.1 Proposed System Behaviour

The proposed Online Quiz System provides a structured platform for conducting multiple-choice quizzes. It is intended to reduce manual effort in quiz management and evaluation by automating question presentation, answer collection, score calculation, and result generation.

3.2 Participant Workflow

Start Quiz → View Question → Select Answer → Navigate Questions → Submit Quiz → Calculate Score → View Score → View Result → Exit Quiz

During submission, the system checks for unanswered questions according to the quiz rules. If unanswered questions are found, the system displays a warning before the quiz is submitted, as applicable.

3.3 Administrator Workflow

Login → Validate Login Credentials → Verify Administrator Access → View Questions → Add / Edit / Delete Question → Exit Administrative Session

The administrator can add, edit, or delete questions only after successful authentication. The system validates question information before storing changes and maintains data consistency.

4. UML Use Case Model

4.1 Participant Use Cases

Use Case	Description	Related Requirement
Start Quiz	Allows the participant to start an available quiz.	FR-01
View Question	Displays multiple-choice questions along with their answer options.	FR-02
Select Answer	Allows the participant to select an answer from the available options.	FR-03
Navigate Questions	Allows the participant to move between the available quiz questions.	FR-04
Submit Quiz	Allows the participant to submit the completed quiz.	FR-05
View Score	Displays the score calculated from the submitted answers.	FR-06
View Result	Displays the final quiz result after submission.	FR-07
Exit Quiz	Allows the participant to leave the quiz.	FR-08

4.2 Administrator Use Cases

Use Case	Description	Related Requirement
Login	Allows the administrator to access administrative functions using valid credentials.	FR-09
View Questions	Allows the administrator to view the questions stored in the system.	FR-13
Add Question	Allows the administrator to add a new multiple-choice question.	FR-10
Edit Question	Allows the administrator to modify an existing question.	FR-11
Delete Question	Allows the administrator to delete an existing question.	FR-12

4.3 Include Relationships

Main Use Case	Included Use Case	Reason
Select Answer	Validate Answer Input	The selected answer must be one of the options provided for the question.
Submit Quiz	Validate Unanswered Questions	The system checks whether any questions remain unanswered according to the quiz

		rules.
Submit Quiz	Calculate Score	The system calculates the score using the submitted answers and stored correct answers.
View Result	Display Score	The participant's score is displayed as part of the quiz result.
Login	Validate Login Credentials	The administrator's credentials must be checked before access is granted.
Add Question	Validate Question Input	The new question and its answer options must be checked before storage.
Edit Question	Validate Question Input	The modified question information must be checked before storage.
Add Question	Verify Data Integrity	The system checks that adding a question does not create inconsistent data.
Edit Question	Verify Data Integrity	The system checks that editing a question does not create inconsistent data.
Delete Question	Verify Data Integrity	The system checks that deleting a question does not leave inconsistent data.

4.4 Extend Relationships

Base Use Case	Extending Use Case	Condition
Submit Quiz	Display Unanswered Question Warning	Extends the submission process when unanswered questions are found.
Login	Display Login Error	Extends the login process when the entered credentials are incorrect.
Add Question	Display Validation Error	Extends the add-question process when the entered information is invalid.
Edit Question	Display Validation Error	Extends the edit-question process when the entered information is invalid.
Delete Question	Display Operation Error	Extends the delete-question process when the operation cannot be completed.

5. Security Design Information Available in the Source

The source document identifies authentication, authorization, session management, input handling and data protection as security validation areas. It does not provide a separate security architecture.

Security Requirement ID	Requirement
VFR-03	The system shall not allow an unauthorized user to access administrator functions.
VFR-04	The system shall reject an administrator login attempt when the entered credentials are incorrect.
VFR-09	The system shall prevent unauthorized users from accessing an administrator session.

6. Design Information Available in the Source

6.1 Question and Answer Validation

- The system shall reject an answer that is not one of the options provided for that question.
- The system shall identify unanswered questions before the quiz is submitted, according to the quiz rules.
- The system shall not accept incomplete or invalid question information from the administrator.
- The system shall check the question and answer data before storing it in the database.

6.2 Score Calculation and Result Display

The system shall calculate the user's score based on the answers submitted and the correct answers stored in the system. The system shall display the user's score and result after the quiz is submitted.

6.3 Error Handling

The system shall show a suitable error message when the user enters invalid data or performs an invalid operation.

- Display an unanswered-question warning when unanswered questions are found.
- Display a login error when the entered administrator credentials are incorrect.
- Display a validation error when question information is invalid.
- Display an operation error when a delete-question operation cannot be completed.

7. Traceability to Requirements

System Element / Use Case	Related Requirement
Start Quiz	FR-01
View Question	FR-02
Select Answer	FR-03
Navigate Questions	FR-04
Submit Quiz	FR-05
Calculate Score / View Score	FR-06
View Result	FR-07
Exit Quiz	FR-08
Administrator Login	FR-09
Add Question	FR-10
Edit Question	FR-11
Delete Question	FR-12
View Questions	FR-13

8. Source Coverage

The supplied Phase 1 PDF contains actor descriptions, use cases, participant and administrator workflows, use-case include and extend relationships, functional requirements, validation/verification requirements and a use-case diagram section. It does not contain a component diagram, a named architecture pattern, sequence diagrams, API specifications or a separate security architecture. These items have not been created or inferred because the instruction was to use only information in the supplied PDF.

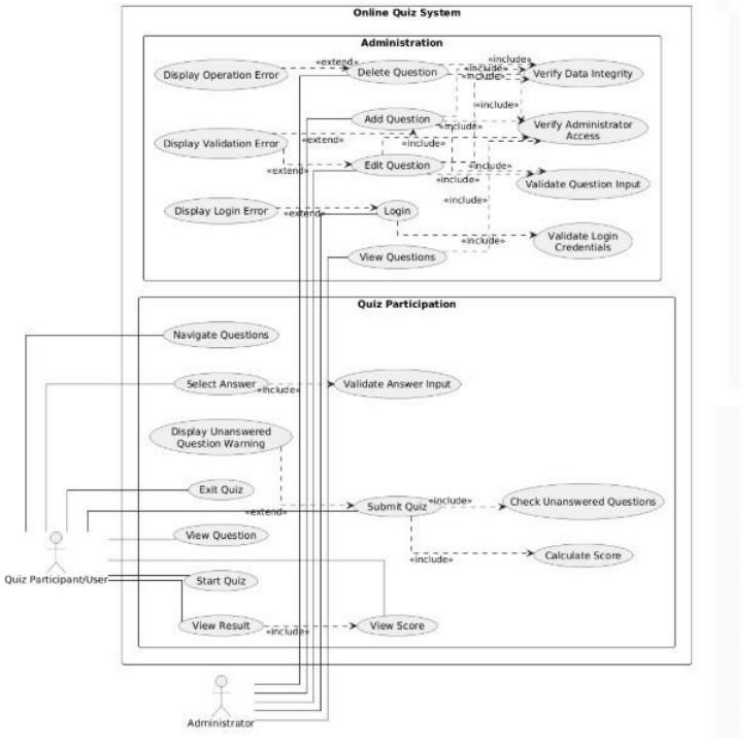
Appendix A. Use Case Diagram Page from the Source PDF

The following page image is included from the supplied PDF without redesign.

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Add Question	Display Validation Error	Extends the add-question process when the entered information is invalid.
Edit Question	Display Validation Error	Extends the edit-question process when the entered information is invalid.
Delete Question	Display Operation Error	Extends the delete-question process when the operation cannot be completed.

These relationships represent the validation and error-handling behaviour described in the Validation/Verification Requirements. They do not introduce additional major actors or unrelated system features.

SECTION 7 — UML USE CASE DIAGRAM



7.1
Description of the Use Case Diagram

The UML Use Case Diagram represents the interactions between the two main actors and the Online Quiz System.
The Quiz Participant interacts with the system to start a quiz, view multiple-choice questions,